

Q1: What is a generic class? Why use generics?

Q2: Write a generic class `Container<T>` with Add and Get methods.

Q3: What are multiple type parameters? Write `Pair<TKey, TValue>`.

Q4: What is a generic method? Write `Swap<T>` method.

Q5: Write a generic method `FindMax<T>` that finds maximum value

Q6: What is a generic interface? Write `IRepository<T>`.

Q7: What is the 'struct' constraint? Write an example.

Q8: What is the 'class' constraint? Write an example.

Q9: What is the 'new()' constraint? Write an example.

Q10: What is the interface constraint? Write an example.

Q11: What is the base class constraint? Write an example.

Q12: How do you apply multiple constraints? Write an example.

Q13: What does the 'default' keyword do in generics?

Q14: Write a `SafeList<T>` that returns default when the index is invalid.

Q15: What is covariance? Explain the 'out' keyword.

Q16: What is contravariance? Explain the 'in' keyword.

Q17: What is the difference between covariance and contravariance?

Q18: How do static members work in generic types?

Q19: How can you inherit from a generic class?

Q20: Complete Exercise - Create a generic `Cache<TKey, TValue>` with Add, Get, Remove, Contains, and expiration support.